

Forum

A role for epigenetics in rapid adaptation

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Epigenetic variation may be key to rapid adaptation and can interact with genetic change during evolutionary rescue. However, key questions remain, including whether epigenetic shifts precede genetic change, persist long-term, or influence fitness. We propose that temporal sampling across multigenerational studies is the most promising means to close these gaps.

What is the role of epigenetics in evolutionary rescue?

Epigenetic mechanisms are frequently invoked as contributors to rapid adaptation and evolutionary rescue under global change conditions, yet direct tests of this idea remain scarce. Epigenetic changes are modifications to DNA, including those that are not inherited, which can impact gene expression without changing the underlying sequence itself and thus may act as a form of phenotypic plasticity—the ability of a genotype to express multiple phenotypes [1]. Because these epigenetic shifts can occur faster than genetic changes, epigenetics may enable adaptive responses to environmental change, particularly when environmental change outpaces generation time. Indeed, in some contexts, reducing the available epigenetic variation can reduce the degree of adaptation; experimental work in algae has shown that the ability to produce and transmit epigenetic variation can shape a population's ability to adapt [2].

While numerous authors, including ourselves, have argued for the importance

of epigenetics in rapid adaptation [3], there have been few direct tests of this idea. Many studies have occurred in plants and have shown, for example, that epigenetic changes are likely linked to phenotypic diversity and adaptation to local environments [4]. Plant studies may be more common than those in animals, in part due to their short generation time, potential for clonal propagation, and established experimental protocols, but also because environmentally driven epigenetic alterations in plant somatic cells might be more easily transmitted to the germline and passed on to offspring as compared to animals [5].

There have been fewer studies in animal systems, perhaps partly because the experimentally amenable *Drosophila* only sparsely uses methylation. Of existing animal studies, many are correlational in nature. For example, on relatively short timeframes, our experimental evolution work in copepods has shown that genetic and epigenetic changes over 25 generations are complementary during evolutionary rescue: genomic regions exhibiting consistent epigenetic divergence had $2.5 \times$ lower genetic divergence than the rest of the genome [3]. Across longer timescales, comparing locally adapted populations, genetic variation can explain a significant proportion of epigenetic variation [6]. However, in most cases, there is a substantial proportion of epigenetic variation that is unexplained by genetic factors and is likely driven by the environment [7]. Finally, while transgenerational epigenetic inheritance is well established, the role of epigenetic changes during evolutionary rescue remains poorly resolved, particularly in nonmodel animal systems under environmental stress [8]. Important questions remain about the timing, fitness consequences, and long-term fate of these dynamics (Figure 1).

There are critical tests that could address the role of epigenetics in rapid adaptation. In this forum, we focus on nonmodel

animal systems, as we believe they are underrepresented in the literature and are ecologically essential. Further, when referring to epigenetics, we are predominantly referring to DNA methylation, as it is easy to measure across diverse organisms and has been widely studied. However, other forms of epigenetics, such as chromatin modification, exist and are important; the arguments presented here extend to these modifications as well. Overall, we encourage increased use of experimental systems, as well as temporal datasets and recently established populations, such as invasive species or domestication events, to address the gaps in knowledge surrounding the role of epigenetics in rapid adaptation and evolutionary rescue.

Do epigenetic changes precede genetic changes?

If epigenetic mechanisms contribute to rapid responses to global change, these changes may arise first, after which genetic changes follow (Figure 1A). That is, epigenetic changes are plastic and facilitate initial survival in the novel environment, after which evolution can follow, that is, the Baldwin effect. Alternatively, genetic and epigenetic changes may arise in parallel, both contributing to rapid responses (Figure 1B). However, the order in which these changes occur is unclear; few studies leverage time series data to address this question. For example, in great tits (*Parus major*) subjected to genomic selection for egg-laying date, both types of variants were differentiated between selection lines [9]. Our own work with copepods has shown similar complementary shifts [3]. However, these studies have only tracked the beginning and end, not the intermediate generations. A full understanding of the order of establishment and interactions between genetic and epigenetic changes could be addressed using fine-scale time series observations. While these temporal data would help resolve the role of epigenetics in rapid responses

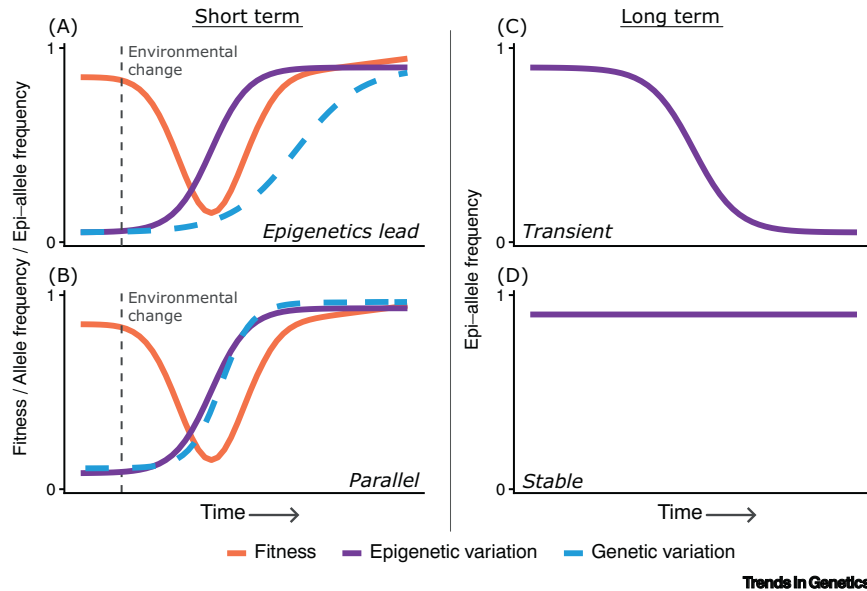


Figure 1. Outstanding hypotheses for the short- and long-term dynamics of genetic and epigenetic changes during and after rapid adaptation and evolutionary response to environmental change. For short-term responses across only a few generations, we hypothesize that (A) epigenetic changes may drive the immediate response and allow for the initial fitness recovery, after which genetic change follows. Conversely, (B) genetic and epigenetic changes may occur in parallel. Over the long-term responses, (C) epigenetic changes may be transient, returning to baseline following the adaptation and recovery of the population. Alternatively, (D) epigenetic changes may be stable across time, due to either continued selective pressures or fixation via genetic mechanisms.

to environmental change, neither pattern necessarily implies causation.

What is the long-term fate of early epigenetic changes?

In addition to the order of establishment of genetic and epigenetic changes, the long-term stability of epigenetic changes remains unclear. We propose two hypotheses for the fate of these early changes. First, if epigenetic changes are acting as a form of plasticity, they could undergo genetic assimilation, being replaced with genetic mechanisms, which has recently been demonstrated in medaka fish [10]. In this scenario, epigenetic changes would be transient and dissipate with time (Figure 1C). Alternatively, epigenetic changes may be under selection and remain stable over long periods due to either maintenance of the plastic epigenetic state under stable conditions, transgenerational inheritance, or fixation

via genetic effects (Figure 1D). This has been observed in the bacterium *Ralstonia pseudosolanacearum* under experimental adaptation to a host plant [11]. In this system, a methylation change that directly impacted gene expression and was linked to adaptation arose early in the experiment and persisted for 300 generations. It is plausible that similar long-term stable epialleles may occur in animal systems.

Can early epigenetic changes be linked to fitness?

Central to the idea that epigenetic changes facilitate rapid adaptation is the notion that these changes improve fitness, that is, survival and reproductive success. These changes can consist of transient (inducible) modifications representing plasticity and stable (noninducible) modifications that are driven by genetics. Here, it is important to distinguish between loci that are associated with fitness

improvements versus those that are causal in improving fitness. In the context of local adaptation, stable and divergent methylation patterns between locally adapted populations suggest that these patterns have been shaped by selection and play a role in adaptation and fitness. For example, sticklebacks locally adapted to low- versus high-salinity environments show stable, noninducible differentiation in methylation [12]. In other cases, epigenetic variation can explain phenotypic variation: in Pacific oysters, approximately 25% of the phenotypic variation in resistance to mortality syndrome was explained by epigenetics, while genetic variation explained just approximately 14% [13]. More generally, numerous studies, including our own [3], have shown that epigenetic variation is correlated with gene expression and potentially phenotypic variation, at least at the molecular level. However, demonstrations of epigenetic variation as the cause of variation in fitness, rather than merely a correlation, have been rare and remain a critical gap.

Multigeneration strategies for connecting genetic and epigenetic variation

We believe that multigenerational time series provide a promising route to understanding the dynamics between genetic and epigenetic changes. These data would allow one to directly address the order of establishment of genetic versus epigenetic changes, the long-term fate of epigenetic changes, and potentially reveal links between epigenetics and fitness. There are three most obvious sources for these data: experimental evolution, invasive species, and long-term monitoring.

Experimental evolution provides the most controlled environment to quantify these dynamics and is the approach we have chosen in the past [3]. With this method, one can closely control the selection pressures and directly track genetic and epigenetic changes at each generation,

allowing precise quantification of the emergence of each. Similarly, sampling during the process of domestication could provide temporal data on genetic and epigenetic changes, for example, comparing founding populations versus domesticated lineages [14].

Recently established invasive species provide an alternative route for temporal samples. These invasive species provide a natural, albeit unwanted, ‘experiment’ where populations are responding to and typically thriving under novel, natural conditions. Invasive species commonly establish with little evidence of adaptive genetic divergence [15], suggesting that plastic and potentially epigenetic mechanisms may play a role. Tracking temporal genetic and epigenetic changes could reveal important dynamics during adaptation to a new environment.

Finally, long-term collections, such as museums or fisheries monitoring, may also be a viable source for time series data. In certain regions, environments have changed considerably in recent years, spanning multiple generations for many species. Where collections are consistent in space and stored appropriately for epigenomics, these data could provide a valuable source of information for populations undergoing adaptation. These collections may also provide the opportunity to quantify changes across much longer timescales than are feasible with an experimental approach.

Across these methods, a primary challenge is disentangling correlation from causation. One solution is to remove genetic variation by using clonal or inbred lines. Alternatively, methylation can be experimentally removed globally [2] using genetic or chemical means, or at specific

loci, for example, with CRISPR-based approaches, and the fitness impacts quantified. However, these approaches have limitations, such as DNA damage and genome instability from chemical demethylation, and off-target methylation changes and DNA damage from CRISPR, but they could be useful tools toward validating causation rather than correlation.

Concluding remarks

Epigenetic mechanisms are plausible, important, but under-tested contributors to resilience, rapid adaptation, and evolutionary rescue under changing environments. The challenge is to understand the functional dynamics of genetic and epigenetic variation, that is, the order, persistence, and fitness consequences of epigenetic changes. As a field, we must test when, how, and in which systems epigenetic mechanisms matter for rapid adaptation and rescue. In our opinion, the best way to accomplish this is to utilize temporally resolved, fitness-linked, multi-generational studies. By leveraging these time series data and experimental approaches, we will begin to uncover the nuanced roles and mechanisms of epigenetic and genetic changes in evolutionary rescue.

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Declaration of interests

The authors declare no competing interests.

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References

- Bogan, S.N. and Yi, S.V. (2024) Potential role of DNA methylation as a driver of plastic responses to the environment across cells, organisms, and populations. *Genome Biol. Evol.* 16, evae022
- Kronholm, I. et al. (2017) Epigenetic and genetic contributions to adaptation in *Chlamydomonas*. *Mol. Biol. Evol.* 34, 2285–2306
- Brennan, R.S. et al. (2025) Complementary genetic and epigenetic changes facilitate rapid adaptation to multiple global change stressors. *Proc. Natl. Acad. Sci. U. S. A.* 122, e2422782122
- Shahzad, Z. et al. (2025) Gene body methylation regulates gene expression and mediates phenotypic diversity in natural *Arabidopsis* populations. *Nat. Plants* 11, 2084–2099
- Guo, H. et al. (2026) Testing Weismann’s germ plasm theory in *Arabidopsis*. *Curr. Biol.* 36, 1918–1931.e6
- Silliman, K. et al. (2023) Epigenetic and genetic population structure is coupled in a marine invertebrate. *Genome Biol. Evol.* 15, evad013
- Johnson, K.M. and Kelly, M.W. (2020) Population epigenetic divergence exceeds genetic divergence in the eastern oyster *Crassostrea virginica* in the northern Gulf of Mexico. *Evol. Appl.* 13, 945–959
- Shahmohammadloo, R.S. et al. (2025) Transgenerational epigenetic inheritance increases trait variation but is not adaptive. *Evolution* 79, 1033–1043
- Lindner, M. et al. (2024) Genetic and epigenetic differentiation in response to genomic selection for avian lay date. *Evol. Appl.* 17, e13703
- Katsumura, T. et al. (2026) DNA methylation site loss for plasticity-led novel trait genetic fixation. *Proc. Natl. Acad. Sci. U. S. A.* 123, e2534817123
- Gopalan-Nair, R. et al. (2024) Changes in DNA methylation contribute to rapid adaptation in bacterial plant pathogen evolution. *PLoS Biol.* 22, e3002792
- Heckwolf, M.J. et al. (2020) Two different epigenetic information channels in wild three-spined sticklebacks are involved in salinity adaptation. *Sci. Adv.* 6, eaaz1138
- Gawra, J. et al. (2023) Epigenetic variations are more substantial than genetic variations in rapid adaptation of oyster to Pacific oyster mortality syndrome. *Sci. Adv.* 9, eadh8990
- Griffiths, J.S. et al. (2026) Highly and lowly domesticated endangered fish from a conservation hatchery diverge in their thermal physiology, transcriptome, and methylome. *Evol. Appl.* 19, e70220
- Bentley, B.P. et al. (2024) Successful invasion into new environments without evidence of rapid adaptation by a predatory marine gastropod. *Mol. Ecol.* 33, e17575